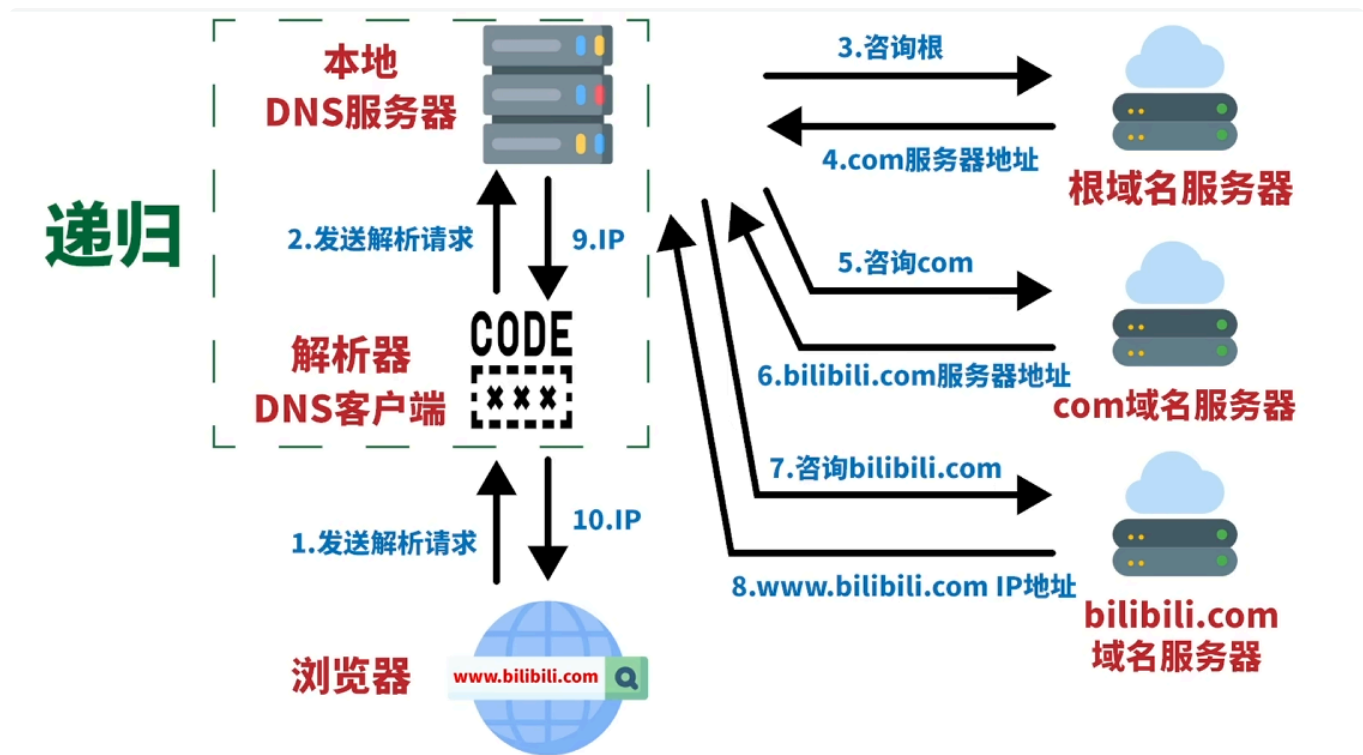


递归查询



具体过程

假设本地DNS服务器没有任何根域名服务器的域名和地址：

nslookup工具进行查询

获取根服务器IP

set type=ns： 查询类型为ns也就是名称域名服务器name server，查询该域名的 **权威 DNS 服务器** 地址

∴ 查询根服务器的域名输入一个.表示根

```

└─# nslookup
> set type=ns
> .
Server:      192.168.230.2
Address:     192.168.230.2#53

Non-authoritative answer:
.           nameserver = h.root-servers.net.
.           nameserver = f.root-servers.net.
.           nameserver = b.root-servers.net.
.           nameserver = l.root-servers.net.
.           nameserver = c.root-servers.net.
.           nameserver = e.root-servers.net.
.           nameserver = d.root-servers.net.
.           nameserver = g.root-servers.net.
.           nameserver = a.root-servers.net.
.           nameserver = j.root-servers.net.
.           nameserver = i.root-servers.net.
.           nameserver = k.root-servers.net.
.           nameserver = m.root-servers.net.

Authoritative answers can be found from:

```

这里返回了非权威的名称域名服务器结果，这里读取的是本地缓存
得到了域名还需要IP才能通信

set type=a：设置ipv4格式

b.root-servers.net.：选择其中一个根服务器进行IP查询

```

> set type=a
> b.root-servers.net.
Server:      192.168.230.2
Address:     192.168.230.2#53

Non-authoritative answer:
Name:       b.root-servers.net
Address:    170.247.170.2
>

```

得到这个根的IP地址（非权威）

获取顶级域名com的IP

server 170.247.170.2：与这个根域名服务器进行通信

set type=ns：查询类型为ns也就是名称域名服务器name server

com.: 像根域名服务器咨询com域名服务器的地址

```
> server 170.247.170.2
Default server: 170.247.170.2
Address: 170.247.170.2#53
> set type=ns
> com.
;; communications error to 170.247.170.2#53: timed out
Server:      170.247.170.2
Address:     170.247.170.2#53

Non-authoritative answer:
*** Can't find com.: No answer

Authoritative answers can be found from:
com      nameserver = a.gtld-servers.net.
com      nameserver = b.gtld-servers.net.
com      nameserver = c.gtld-servers.net.
com      nameserver = d.gtld-servers.net.
com      nameserver = e.gtld-servers.net.
com      nameserver = f.gtld-servers.net.
com      nameserver = g.gtld-servers.net.
com      nameserver = h.gtld-servers.net.
com      nameserver = i.gtld-servers.net.
com      nameserver = j.gtld-servers.net.
com      nameserver = k.gtld-servers.net.
com      nameserver = l.gtld-servers.net.
com      nameserver = m.gtld-servers.net.
```

顶级域名服务器的域名（权威回答）

```
a.gtld-servers.net      internet address = 192.5.6.30
a.gtld-servers.net      has AAAA address 2001:503:a83e::2:30
b.gtld-servers.net      internet address = 192.33.14.30
b.gtld-servers.net      has AAAA address 2001:503:231d::2:30
c.gtld-servers.net      internet address = 192.26.92.30
c.gtld-servers.net      has AAAA address 2001:503:83eb::30
d.gtld-servers.net      internet address = 192.31.80.30
d.gtld-servers.net      has AAAA address 2001:500:856e::30
e.gtld-servers.net      internet address = 192.12.94.30
e.gtld-servers.net      has AAAA address 2001:502:1ca1::30
f.gtld-servers.net      internet address = 192.35.51.30
f.gtld-servers.net      has AAAA address 2001:503:d414::30
```

gtld: generic top leve domain（通用顶级域名）

顶级域名服务器域名所对应的IP地址（权威回答）

获取bilibili.com域名服务器IP

server 192.33.14.30:

set type=ns:

bilibili.com.: 向一个顶级域名服务器的地址192.33.14.30咨询bilibili.com域名服务器的地址

```
> server 192.33.14.30
Default server: 192.33.14.30
Address: 192.33.14.30#53
> set type=ns
> bilibili.com.
Server:      192.33.14.30
Address:     192.33.14.30#53

Non-authoritative answer:
*** Can't find bilibili.com.: No answer

Authoritative answers can be found from:
bilibili.com      nameserver = ns3.dnsv5.com.
bilibili.com      nameserver = ns4.dnsv5.com.
```

bilibili域名服务器有两个 (ns3.dnsv5.com.与ns4.dnsv5.com.) (权威回复)

```
ns3.dnsv5.com      internet address = 1.12.0.17
ns3.dnsv5.com      internet address = 1.12.0.18
ns3.dnsv5.com      internet address = 1.12.14.17
ns3.dnsv5.com      internet address = 1.12.14.18
ns3.dnsv5.com      internet address = 101.227.168.52
ns3.dnsv5.com      internet address = 108.136.87.44
ns3.dnsv5.com      internet address = 163.177.5.52
ns3.dnsv5.com      internet address = 220.196.136.52
ns3.dnsv5.com      has AAAA address 2402:4e00:1470:2::f
ns3.dnsv5.com      internet address = 35.165.107.227
ns4.dnsv5.com      internet address = 1.12.0.16
ns4.dnsv5.com      internet address = 1.12.0.19
ns4.dnsv5.com      internet address = 1.12.14.16
ns4.dnsv5.com      internet address = 1.12.14.19
ns4.dnsv5.com      internet address = 117.135.128.152
ns4.dnsv5.com      internet address = 124.64.205.152
ns4.dnsv5.com      internet address = 13.37.58.163
ns4.dnsv5.com      has AAAA address 2402:4e00:111:fff::8
ns4.dnsv5.com      internet address = 49.7.107.152
```

两个域名对应多个IP地址

获取www.bilibili.com子域名的IP

server 101.227.168.52: 与ns3.dnsv5.com.的其中一个地址进行通信

set type=a: a 代表 Address (地址) 记录

www.bilibili.com : 向权威服务器咨询www.bilibili.com域名服务器的IPv4地址

```
> server 101.227.168.52
Default server: 101.227.168.52
Address: 101.227.168.52#53
> set type=a
> www.bilibili.com
Server:      101.227.168.52
Address:     101.227.168.52#53

www.bilibili.com      canonical name = a.w.bilicdn1.com.
```

这里canonical name = a.w.bilicdn1.com.说明www.bilibili.com 会跳转到a.w.bilicdn1.com.这里的a.w.bilicdn1.com.其实是一个cdn节点域名

a.w.bilicdn1.com. : 查询该cdn节点的IP地址

```
> a.w.bilicdn1.com.
Server:          101.227.168.52
Address: 101.227.168.52#53

Name:   a.w.bilicdn1.com
Address: 117.23.60.15
Name:   a.w.bilicdn1.com
Address: 119.84.174.66
Name:   a.w.bilicdn1.com
Address: 119.84.174.67
Name:   a.w.bilicdn1.com
Address: 119.84.174.68
Name:   a.w.bilicdn1.com
Address: 119.84.174.96
Name:   a.w.bilicdn1.com
Address: 119.84.174.97
Name:   a.w.bilicdn1.com
Address: 222.210.39.57
Name:   a.w.bilicdn1.com
Address: 222.210.39.58
Name:   a.w.bilicdn1.com
Address: 222.210.39.59
Name:   a.w.bilicdn1.com
Address: 116.207.163.63
Name:   a.w.bilicdn1.com
Address: 116.207.163.64
Name:   a.w.bilicdn1.com
Address: 116.207.163.65
Name:   a.w.bilicdn1.com
Address: 116.207.163.66
Name:   a.w.bilicdn1.com
Address: 117.23.60.12
Name:   a.w.bilicdn1.com
Address: 117.23.60.13
Name:   a.w.bilicdn1.com
Address: 117.23.60.14
>
```

流程

1. 查 "." 根域 → 得到 13 台根服务器 (a.root-servers.net ~ m.root-servers.net)
2. 查 "com." 顶级域 → 得到 gtld-servers.net (13 台)
3. 查 "bilibili.com." 权威 NS → 得到 ns3.dnsv5.com / ns4.dnsv5.com (腾讯云 DNSPod)
4. 向权威 NS 查询 "www.bilibili.com" → 得到 CNAME (a.w.bilicdn1.com) → 最终 A 记录 (多个 IP, 包含 117.23.60.15)